

Threat modeling with the PASTA framework

1. Project description

This project involved performing a threat modeling exercise for a mobile marketplace application designed for sneakers buyers and sellers. Using the Process for Attack Simulation and Threat Analysis (PASTA) framework, I evaluated the application's business objectives, technical architecture, attack surface, potential threats, vulnerabilities, and security controls before deployment.

2. Skills demonstrated

- threat modeling
- Risk assessment
- Application security
- Secure Software Development Lifecycle (SSDLC)
- Vulnerability analysis
- Security architecture review
- Risk mitigation planning
- Security control evaluation

3. Key activities

- defined business and security objectives for a customer-facing mobile application
- Identified critical technologies including APIs, PKI, AES encryption, SHA-256, and SQL databases
- Analyzed application data flows and system interactions
- Evaluated potential threats such as SQL Injection and Session Hijacking
- Identified vulnerabilities including improper input handling and API token weakness
- Reviewed attack tree models to understand possible attack paths
- Recommended technical, operational, and administrative security controls.

4. Security controls evaluated

- SHA-256 hashing for sensitive data protection,
- Principle of least privileges (PoLP)
- Password security policies
- Incident response procedures

5. Outcome

This project provided practical experience applying the PASTA threat modeling framework to a real-world business application. It strengthened my ability to identify security risks early in the software development lifecycle, assess attack surfaces, evaluate vulnerabilities, and recommend controls that reduce organizational risk before product release.